

Politecnico di Torino

Master's Degree in Computer Engineering

**Cloud-Native Identity and File Management  
Modernization Using Keycloak OIDC in a Secure  
DevOps Environment**



**Politecnico  
di Torino**

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# Introduction

Between February and August 2026, I completed a six-month internship at Outsight, a company based in Sophia Antipolis, France, specializing in cloud modernization solutions. The internship provided me with the opportunity to work on real-world projects involving cloud-native development, security, and software modernization.

The main objectives of my internship were:

- To create technical diagrams for Outsight's cloud infrastructure.
- To improve the existing software stack by addressing functional and security issues.
- To migrate the user management service to Keycloak OIDC.
- To modernize the file sharing service, making it secure, cloud-native, and integrated with Keycloak OIDC.
- To add new features, such as deployment configuration synchronization.
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- cioaooo padova andiamo cia

# Abstract

This genius thesis investigates ... The main goal of this work is to ... The proposed approach is evaluated through ... The results demonstrate that ...

# Contents

|                                      |           |
|--------------------------------------|-----------|
| <b>Abstract</b>                      | <b>ii</b> |
| <b>1 Introduction</b>                | <b>1</b>  |
| 1.1 Motivation . . . . .             | 1         |
| 1.2 Objectives . . . . .             | 1         |
| 1.3 Thesis Structure . . . . .       | 1         |
| <b>2 Background and Related Work</b> | <b>2</b>  |

# List of Figures

# List of Tables

# Chapter 1

## Introduction

In recent years, computer engineering has played a crucial role in the development of complex and large-scale systems. This thesis focuses on ...

### 1.1 Motivation

The motivation behind this work stems from the need to ...

### 1.2 Objectives

The main objectives of this thesis are:

- to analyze ...
- to design ...
- to evaluate ...

### 1.3 Thesis Structure

This thesis is organized as follows. Chapter 2 introduces the background. Chapter ?? presents the proposed methodology. Chapter ?? discusses the experimental results.

# Chapter 2

## Background and Related Work

This chapter provides an overview of the theoretical background and reviews the most relevant related work.